

Digital Computer Organization

Semester End Examination Solutions (E1PA107T)

Section A: CPU Architecture & Instruction Cycles

1. Define general-purpose registers and list their advantages.

General-purpose registers (GPRs) are high-speed storage locations within the CPU that can store both data and addresses. **Advantages:**

- **Speed:** Faster access compared to main memory.
- **Efficiency:** Reduces the number of memory access cycles.
- **Versatility:** Can be used for various arithmetic, logical, and addressing operations.

2. Describe the sub-cycles of instruction execution: Fetch, Decode, and Execute.

- **Fetch:** The CPU retrieves the instruction from the memory address stored in the Program Counter (PC).
- **Decode:** The Control Unit interprets the opcode of the instruction to determine what operation to perform.
- **Execute:** The CPU carries out the instruction (e.g., performing an addition in the ALU or moving data between registers).

Section B: Memory Systems & Parallelism

3. Explain how cache memory improves CPU performance.

Cache memory is a small, extremely fast memory type situated between the CPU and RAM. It improves performance by storing frequently accessed data and instructions. When the CPU finds needed data in the cache (a "cache hit"), it avoids the much slower process of retrieving it from main memory.

5. Analyze the objectives of parallel processing in a digital system.

Parallel processing involves the simultaneous execution of multiple instructions or tasks to increase computational speed. **Key Objectives:**

- **Throughput:** Increasing the total amount of work done in a given time.
- **Efficiency:** Better utilization of hardware resources.

- **Scalability:** Handling larger datasets by distributing tasks across multiple processors.

Section C: Advanced Computation & Memory Types

7. Evaluate multiplication techniques: Booth's Algorithm vs. Array Multiplier.

For scientific computation involving signed numbers, **Booth's Algorithm** is often preferred as it efficiently handles signed-2's complement multiplication and can reduce the number of partial products. An **Array Multiplier** provides high-speed hardware-based multiplication but becomes significantly more complex and area-consuming as the bit size increases.

OR: Evaluate different types of ROM memories.

ROM (Read-Only Memory) types include **PROM** (Programmable), **EPROM** (Erasable), and **EEPROM** (Electrically Erasable). ROM is non-volatile, making it more reliable than RAM for storing critical firmware (BIOS/UEFI) that must remain intact even when power is lost.